INDIANA UNIVERSITY
BLOOMINGTON

INDIANA UNIVERSITY BICENTENNIAL

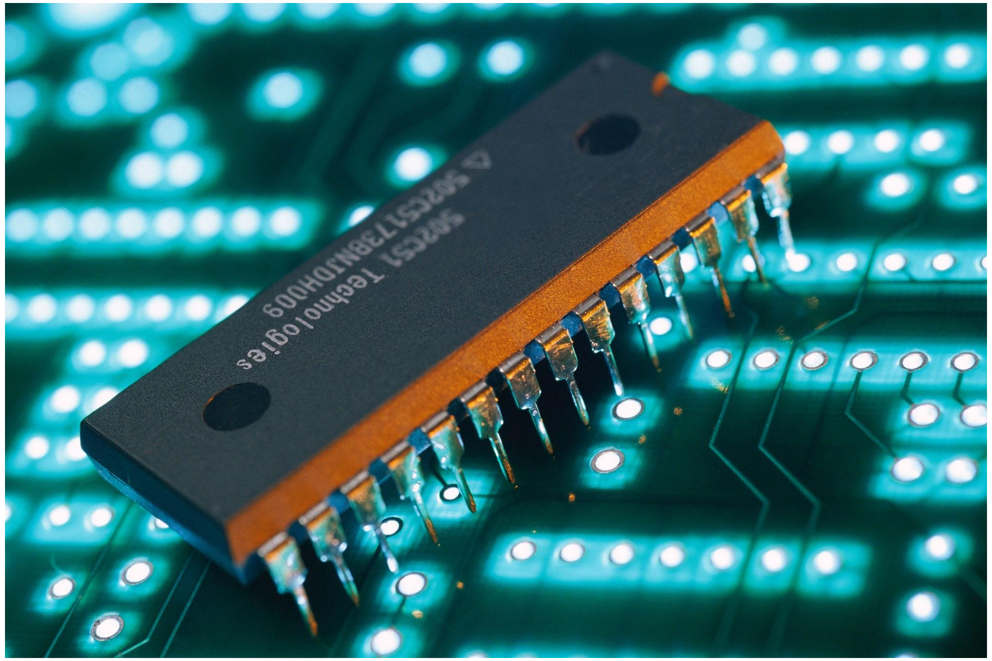
Spring 26 AI on Ramp

Predictive Fault Detection in Semiconductor Manufacturing Using Time-Series Deep Learning

Priyanka Prem Kumar

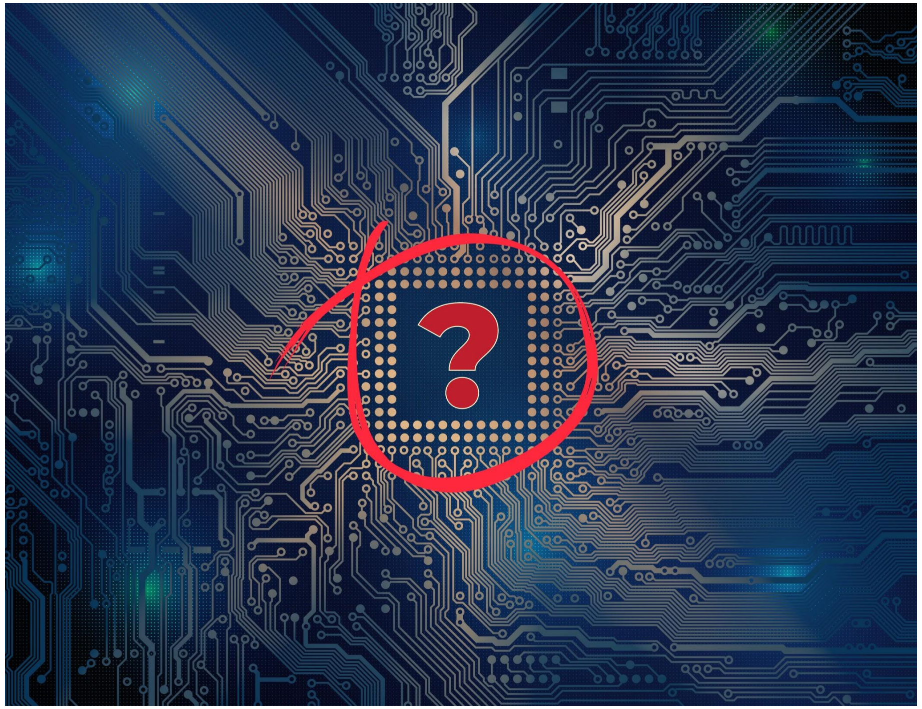
Task 1: Problem Formulation

- **Predictive maintenance in semiconductor manufacturing:** Uses data-driven models to anticipate equipment failures before they occur in complex fabrication systems to reduce downtime and improve yield.
- **High-dimensional sensor data:** Involves large-scale, noisy, and correlated measurements from hundreds of sensors capturing complex semiconductor process conditions.
- **Goal: Early fault prediction:** Aims to detect precursor patterns in time-series sensor data to forecast failures ahead of time for proactive intervention.



Challenge Overview

- **Nonlinear sensor dependencies:** Sensor signals interact in complex, non-linear ways that cannot be captured effectively by simple linear or rule-based models.
- **High-dimensional data:** The dataset contains hundreds of sensor features, making modeling challenging due to sparsity, redundancy, and noise.
- **Rare failure events:** Equipment failures occur infrequently compared to normal operations, leading to severe class imbalance in the data.
- **Limitations of rule-based systems (Sculley et al., 2015):** Traditional rule-based approaches fail to capture evolving temporal patterns and complex relationships in modern industrial data environments.



Dataset: SECOM

- Real semiconductor manufacturing sensor dataset

- High-dimensional features

- Missing values & class imbalance

- Suitable for baseline + temporal modeling

Task 2: Methodology - Modeling Phases

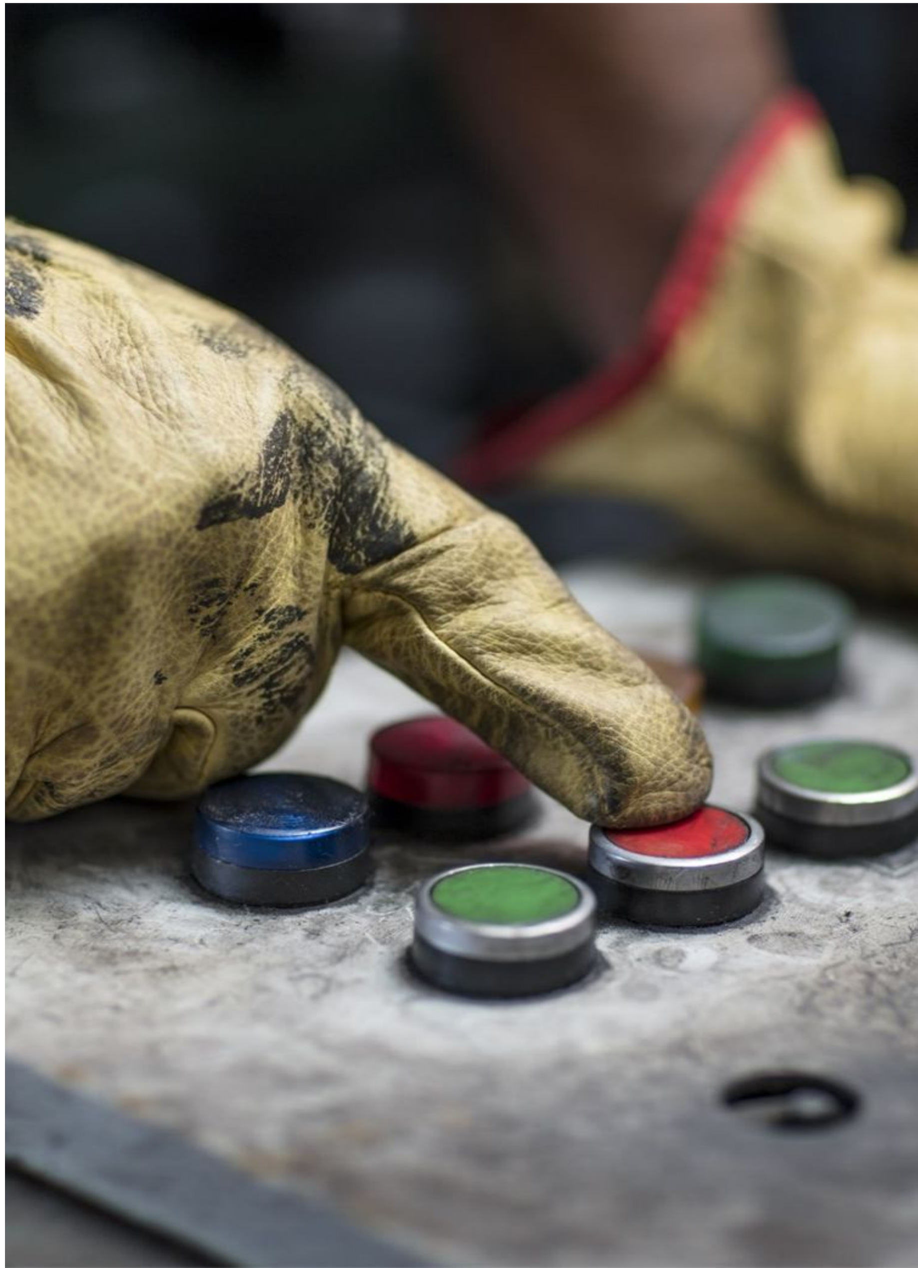
PHASE 1: LOGISTIC REGRESSION (BASELINE)

PHASE 2: LSTM (TEMPORAL LEARNING)

PHASE 3: TRANSFORMER (EARLY WARNING SYSTEM)

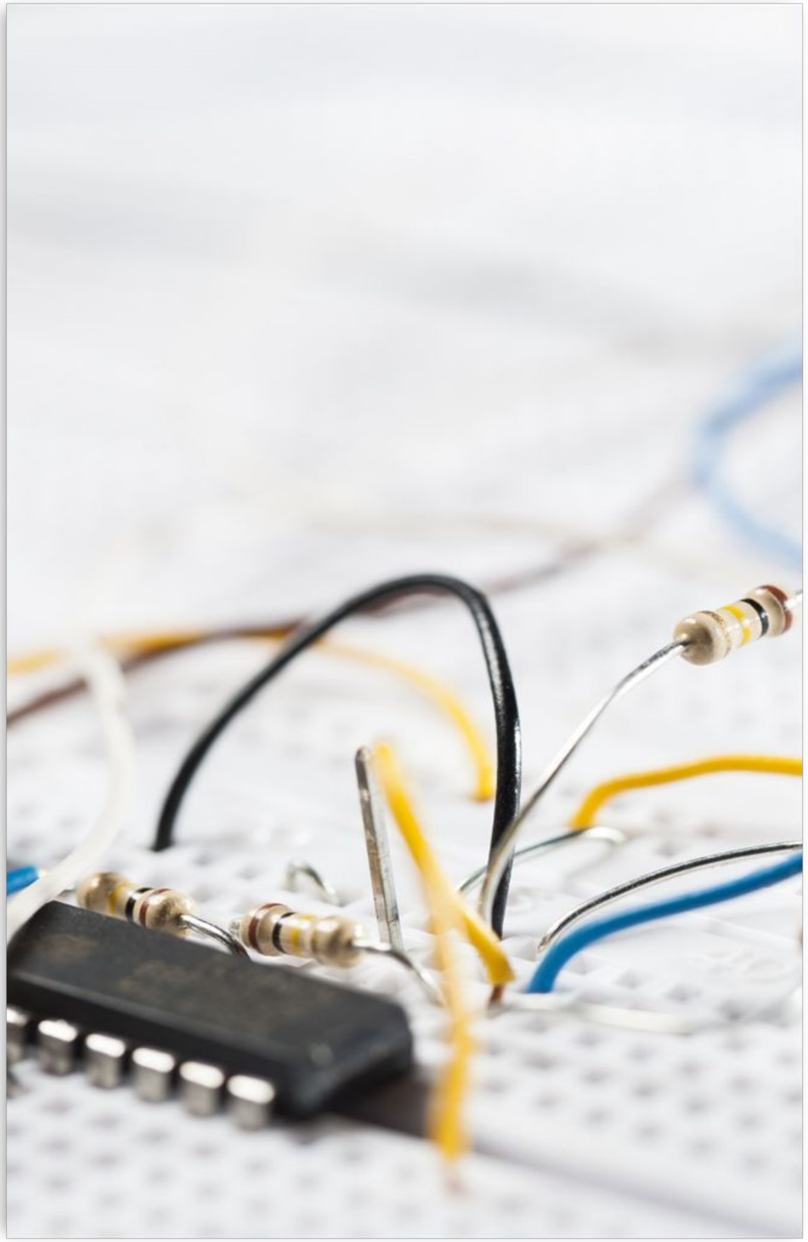
Phase 1: Logistic Regression

- ➢ **Feature cleaning & imputation:** Removes highly incomplete sensor features and fills missing values to ensure a consistent and usable dataset.
- ➢ **Standardization:** Scales all features to a common range to prevent dominance of high-magnitude variables during model training.
- ➢ **Class imbalance handling (class weights):** Assigns higher penalty to minority failure class to ensure the model learns from rare fault events.
- ➢ **Threshold tuning via F1-score:** Optimizes the classification decision threshold to balance precision and recall under severe class imbalance.



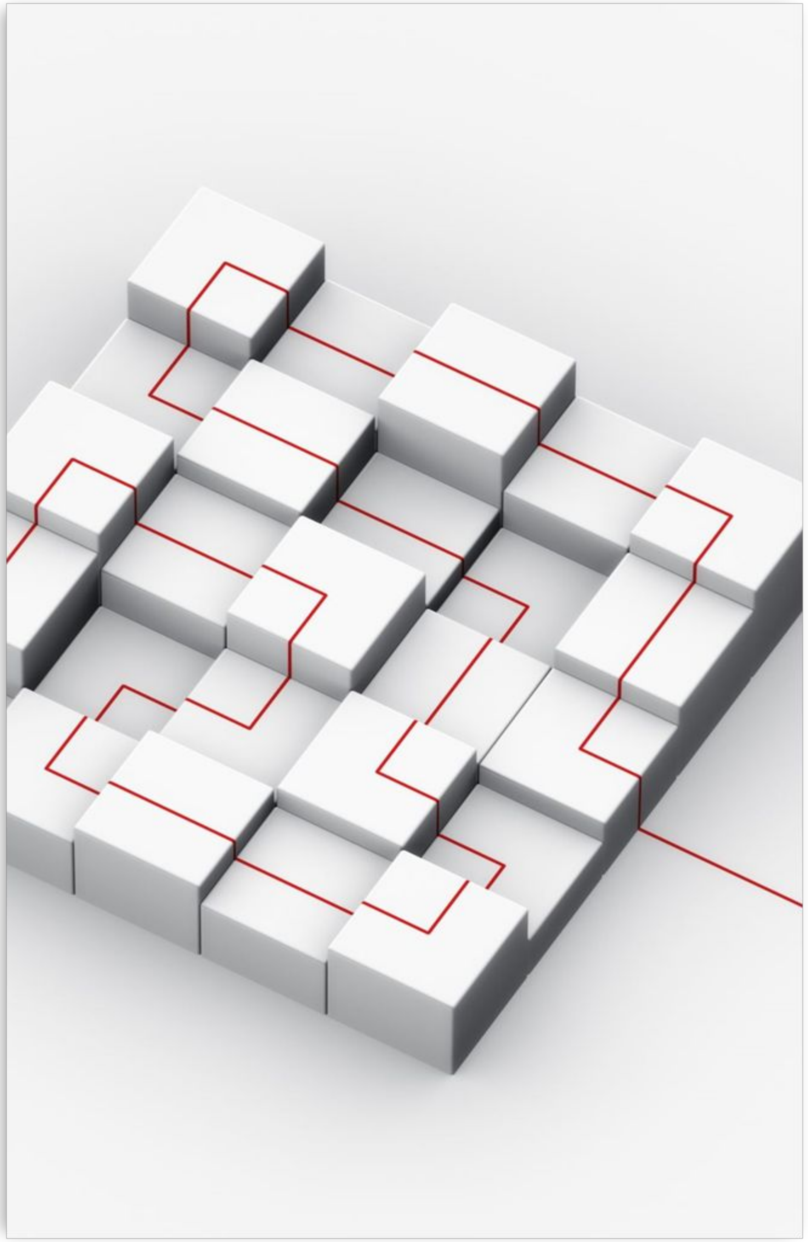
Phase 2: LSTM Model

- **Sequential sensor data:** Organizes sensor readings into time-ordered sequences to capture temporal dependencies leading to failures.
- **Rolling features & deltas:** Enhances signals using moving statistics and first-order differences to highlight trends and sudden changes.
- **Bidirectional LSTM:** Learns temporal patterns by processing sequences in both forward and backward directions for richer context.
- **Early stopping + ROC-AUC monitoring:** Stops training based on validation performance to prevent overfitting while tracking ranking ability across thresholds.



Phase 3: Transformer Model

- **Self-attention architecture:** Uses attention mechanisms to dynamically focus on the most relevant time steps and sensor features in a sequence.
- **Early warning formulation (K-step ahead):** Predicts whether a failure will occur within a future window of K time steps instead of only the next step.
- **Sequence-level representation:** Aggregates information across the entire sequence to create a more robust and noise-resistant prediction.
- **Strong long-range dependency modeling:** Effectively captures relationships between distant time steps that traditional sequential models may miss.



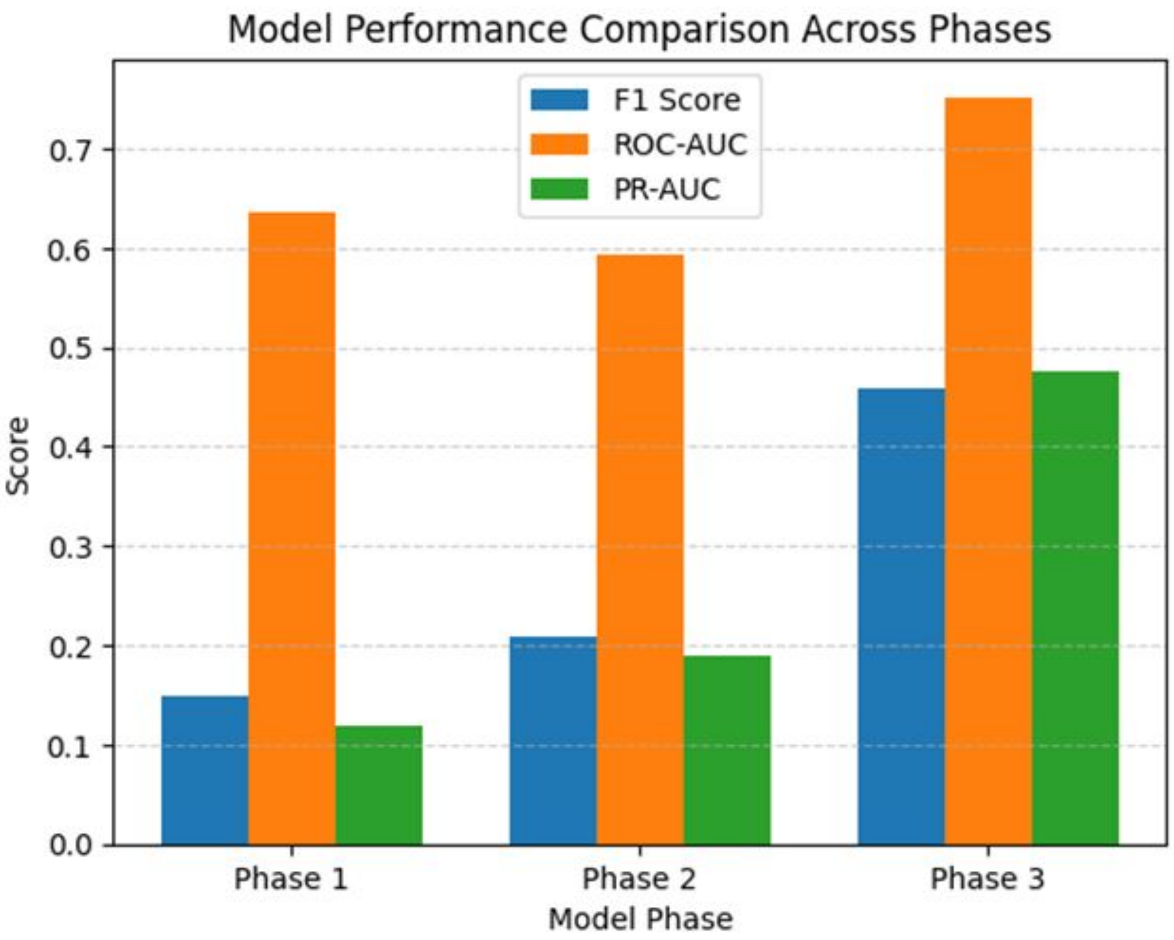
Task 3: Evaluation Metrics

- F1-score (primary)
- ROC-AUC
- PR-AUC (important for imbalance)
- Threshold optimization

Performance Table Summary

PHASE	MODEL	F1 Score	ROC-AUC	PR-AUC
1	Logistic Regression	0.1500	0.6370	0.1191
2	LSTM	0.2095	0.5937	0.1899
3	Transformer	0.4590	0.7519	0.4752

- ➢ **Logistic Regression: baseline performance:**
- ➢ **LSTM: improved recall, unstable ranking:**
- ➢ **Transformer: best overall performance:**



TASK 4: Key Insights

- TEMPORAL MODELING IMPROVES DETECTION

- TRANSFORMER CAPTURES LONG-RANGE DEPENDENCIES

- CLASS IMBALANCE HEAVILY IMPACTS PERFORMANCE

Real-World Impact

- EARLY FAULT DETECTION REDUCES DOWNTIME

- IMPROVES YIELD IN SEMICONDUCTOR MANUFACTURING

- ENABLES PREDICTIVE MAINTENANCE SYSTEMS

Limitations & Future Work

- HYPERPARAMETER TUNING NEEDED
- IMPROVE INTERPRETABILITY
- ADVANCED IMBALANCE TECHNIQUES (FOCAL LOSS)
- DOMAIN-DRIVEN FEATURE ENGINEERING

Conclusion

- ➢ **Transformer-based early warning system performs best:** Delivers the highest performance by capturing complex temporal patterns and predicting failures in advance.
- ➢ **Combining temporal modeling + reformulation is critical:** Integrating sequence learning with an early warning objective significantly improves predictive effectiveness.
- ➢ **Strong industrial applicability:** The approach is well-suited for real-world deployment, enabling proactive maintenance and reducing operational costs.

Related work

- Jesse Davis, J., & Goadrich, M. (2006). The relationship between Precision-Recall and ROC curves. Proceedings of the 23rd International Conference on Machine Learning (ICML).
- Haibo He, H., & Garcia, E. A. (2009). Learning from imbalanced data. IEEE Transactions on Knowledge and Data Engineering, 21(9), 1263–1284.
- Sepp Hochreiter, S., & Jürgen Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation, 9(8), 1735–1780.
- Ashish Vaswani, A., et al. (2017). Attention Is All You Need. Advances in Neural Information Processing Systems (NeurIPS).
- Jay Lee, J., Bagheri, B., & Jin, C. (2016). Introduction to cyber manufacturing. Manufacturing Letters, 8, 11–15.
- McKinsey & Company (2021). The value of predictive maintenance in manufacturing.
- D. Sculley, D., et al. (2015). Hidden technical debt in machine learning systems. Advances in Neural Information Processing Systems (NeurIPS).
- Edward Chow, E., et al. (2001). Time series forecasting for fault detection. IEEE Transactions on Industrial Electronics.
- Tsung-Yi Lin, T.-Y., et al. (2017). Focal Loss for Dense Object Detection. Proceedings of ICCV.

Thank you